

Sensor specifications and datasheet-based selection

Lecture 2 of 16 · 80 minutes · Module A: Foundations

Today one number decides a design — and it is not on any front page.

Last week's muddiest points

Your questions, answered before we start

1

INSTRUCTOR: fill these three in from the exit tickets before class.

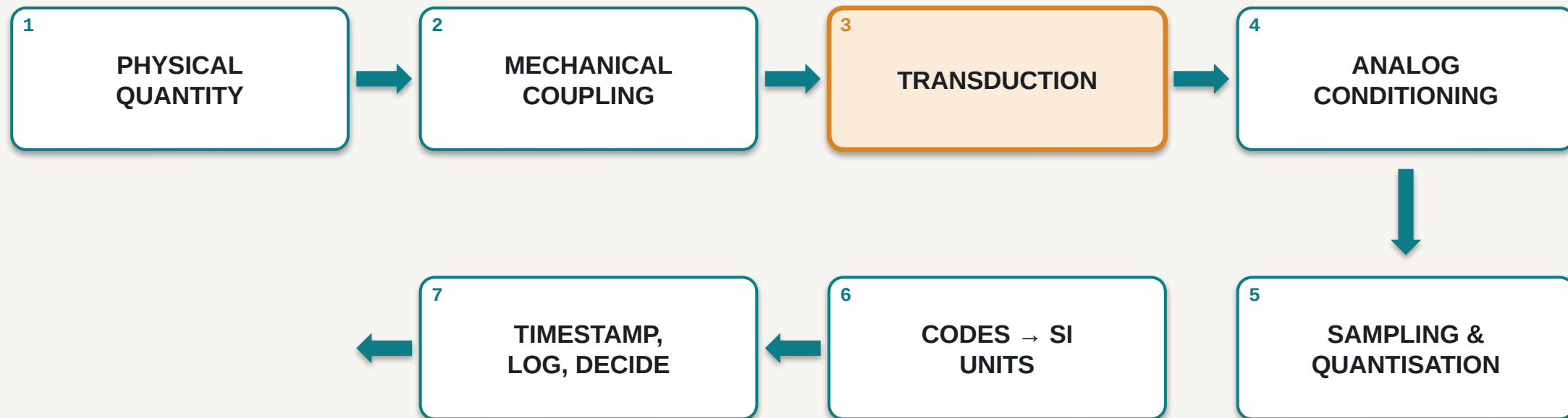
2

3

Refer to concepts, never to students. Ninety seconds total, then move on.

Where are we today?

Retrieval: last week's chain, this week's box



Everything today happens BEFORE you own any hardware. This is the box you choose — and you choose it with arithmetic.

Two front pages

You must order today. Which one?

PART A

Accelerometer, 3-axis

Resolution	16-bit
Full scale	± 2 g
Sensitivity	0.061 mg/LSB
Noise density	100 $\mu\text{g}/\sqrt{\text{Hz}}$
Unit price	€1.80

“HIGH RESOLUTION · LOW NOISE”

PART B

Accelerometer, 3-axis

Resolution	14-bit
Full scale	± 4 g
Sensitivity	0.488 mg/LSB
Noise density	100 $\mu\text{g}/\sqrt{\text{Hz}}$
Unit price	€4.20

The job: report the tilt of a solar-tracker frame to $\pm 0.5^\circ$, outdoors, 0 to 40 °C.

Which part do you specify for a $\pm 0.5^\circ$ tilt measurement?

A Part A — eight times finer resolution, and cheaper

B Part B

C This cannot be decided from front pages alone

Commit. In seventy minutes I will show you that most of this room just chose a part that cannot meet the requirement — and that its own datasheet said so, on page nine.

First, turn the requirement into a number

You cannot budget an error against an angle

$$\text{tilt of } 0.5^\circ \rightarrow \sin(0.5^\circ) \times 1 \text{ g} = \mathbf{8.73 \text{ mg}}$$

8.73 mg is the entire signal we are trying to measure.

Every error term in this lecture gets compared against those 8.73 mg.

Any term larger than that is fatal. Any term far below it is irrelevant — no matter how good it looks on a front page.

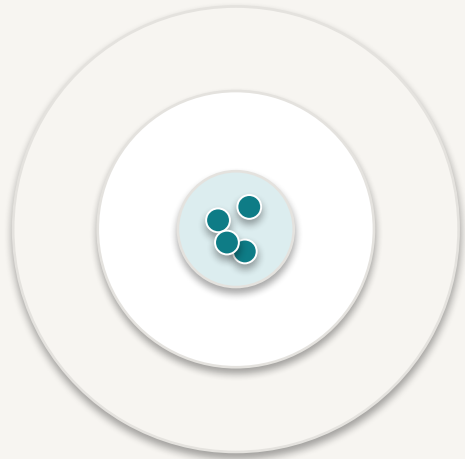
The four words that get confused

Accuracy. Precision. Resolution. Sensitivity.

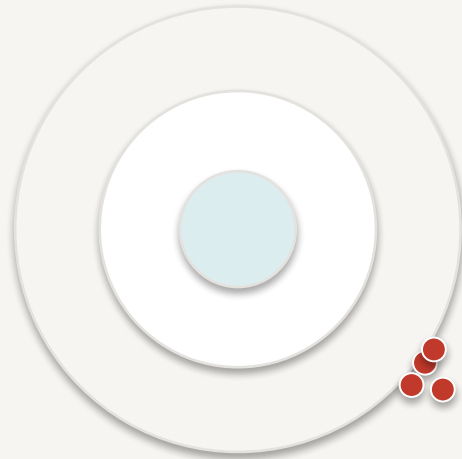
Most engineering arguments about sensors are really arguments about these.

Accuracy is not precision

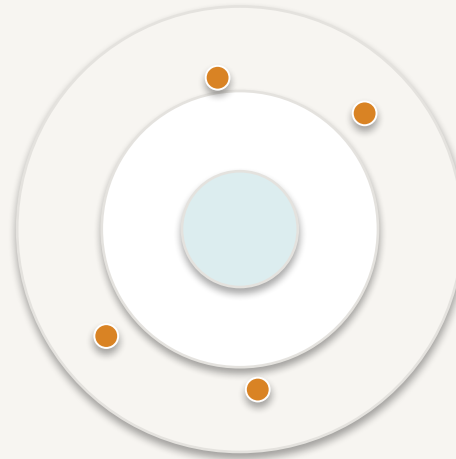
Same target, four different instruments



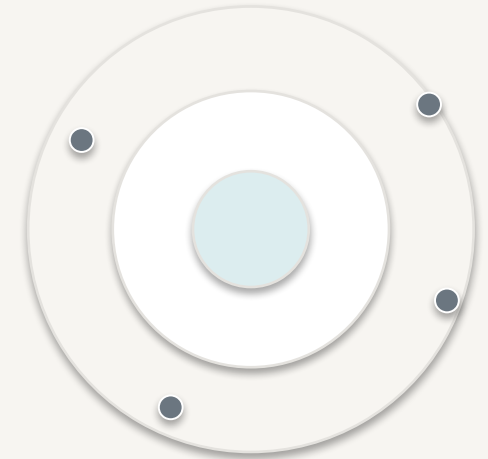
**ACCURATE
AND PRECISE**



**PRECISE, NOT
ACCURATE**



**ACCURATE, NOT
PRECISE**



NEITHER

The centre of the target is the TRUE value. Precision is how tightly the shots group. Accuracy is where the group sits. They are independent — and only one of them is fixable by calibration.

A pressure sensor in a chamber held at a true, constant 1000.0 hPa is read 100 times. Every reading falls between 1012.3 and 1012.5 hPa. The device is:

- A** Accurate but not precise
- B** Precise but not accurate
- C** Both — the spread is only 0.2 hPa
- D** Neither, because the resolution is not stated

Vote alone, then find someone who disagrees.

A pressure sensor in a chamber held at a true, constant 1000.0 hPa is read 100 times. Every reading falls between 1012.3 and 1012.5 hPa. The device is:

A Accurate but not precise

B Precise but not accurate

C Both — the spread is only 0.2 hPa

D Neither, because the resolution is not stated

12.4 hPa of offset, held to 0.2 hPa of scatter. Superb precision, useless accuracy — until somebody calibrates it.

The rule worth memorising

SYSTEMATIC ERROR

is a calibration problem

offset, scale factor — measurable, removable

RANDOM ERROR

is a bandwidth problem

noise — reducible only by averaging, which costs you speed

DRIFT

is a component-selection problem

it moves after you calibrated — so you cannot calibrate it away

This rule is the spine of Lecture 14 and of Laboratories 3 and 5. You will use it in week six.

The vocabulary, sorted

Static: what it does when nothing is moving. Dynamic: what it does when things change.

STATIC CHARACTERISTICS

Range	smallest to largest measurable value
Sensitivity	output change per unit of input
Resolution	smallest distinguishable change
Offset	output when the input is zero
Linearity	deviation from a straight line
Hysteresis	does it matter which way you came?
Repeatability	same input, same output, later

DYNAMIC CHARACTERISTICS

Bandwidth	highest frequency reported faithfully
Response time	time to settle after a step
ODR	data rate — NOT bandwidth
Noise density	noise per $\sqrt{\text{Hz}}$
Group delay	how late the answer arrives
Cross-sensitivity	the axis you did not want
Drift	slow change with time and temperature

What the datasheet actually says

A datasheet is a legal document, not a promise.

The headline is on page one. The truth is in the conditions.

Read the conditions block first

Then, and only then, read the number

Parameter	Min	Typ	Max	Unit	Conditions
Sensitivity	—	0.061	—	mg/LSB	FS = ± 2 g, 25 °C
Zero-g level	—	± 40	—	mg	25 °C, after soldering
Zero-g temp. coefficient	—	± 0.5	—	mg/°C	-40 to +85 °C
Noise density	—	100	—	$\mu\text{g}/\sqrt{\text{Hz}}$	ODR = 200 Hz, BW = 50 Hz
Cross-axis sensitivity	—	± 1	—	%	package level, not board level

► **Every number is “typ”.**

No min, no max. You are being told the average of a production run, not a guarantee.

► **The conditions differ per row.**

Sensitivity at 25 °C. Drift over 125 °C. Noise at one specific bandwidth. They are not comparable as printed.

The hunt

ISM330DHCX datasheet · pairs · 8 minutes · write the page number beside every answer

- 1 Supply voltage range — and the separate I/O supply, if there is one
- 2 Sensitivity at EACH selectable full-scale range
- 3 Zero-offset level AND its temperature coefficient
- 4 Noise density — and the bandwidth it was measured at
- 5 The device-identification register and its expected value
- 6 Whether the headline numbers are typ, min or max — and at what temperature

Number 5 is not academic — you need it at the bench next week to prove your sensor is the sensor you think it is.

**“What is this device's
cross-axis sensitivity after it
has been soldered to my board?”**

It is not in the datasheet. It cannot be. The figure is specified for the packaged die — and mounting stress, solder-joint asymmetry and PCB flex all change it.

Which number was hardest to find?

Hands up. Then tell me why it was hard —
not what the answer was.

The conditions are where the truth lives.

Now compute it

Four terms. One requirement. Twenty minutes.

At the end of this you will know which part you should have chosen.

Noise density becomes noise

...once you choose a bandwidth

$$\text{noise}_{\text{RMS}} = \text{noise density} \times \sqrt{\text{bandwidth}}$$

$$100 \text{ } \mu\text{g}/\sqrt{\text{Hz}} \times \sqrt{50 \text{ Hz}} = 100 \times 7.07 = 707 \text{ } \mu\text{g} = 0.71 \text{ mg}$$

► **The units tell you the answer.**

$\mu\text{g}/\sqrt{\text{Hz}} \times \sqrt{\text{Hz}} = \mu\text{g}$. If your units do not cancel, you have used the wrong number.

► **Bandwidth, not ODR.**

The ODR was 200 Hz. Using $\sqrt{200}$ gives 1.41 mg — a wrong answer that looks right.

► **Halve the bandwidth, gain $\sqrt{2}$.**

Noise is not a property of the sensor alone. It is a property of the sensor and the bandwidth you chose.

Noise density 100 $\mu\text{g}/\sqrt{\text{Hz}}$. You configure ODR = 200 Hz and measurement bandwidth = 50 Hz. What RMS noise appears in your readings?

A $\approx 0.1 \text{ mg}$

B $\approx 0.7 \text{ mg}$

C $\approx 1.4 \text{ mg}$

D $\approx 5 \text{ mg}$

Every wrong answer here is a specific arithmetic mistake — so this is the most useful vote of the day. Take it seriously.

Noise density $100 \mu\text{g}/\sqrt{\text{Hz}}$. You configure $\text{ODR} = 200 \text{ Hz}$ and measurement bandwidth $= 50 \text{ Hz}$. What RMS noise appears in your readings?

A $\approx 0.1 \text{ mg}$

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$100 \mu\text{g}/\sqrt{\text{Hz}} \times \sqrt{50 \text{ Hz}} = 707 \mu\text{g} = 0.71 \text{ mg}$. Now compare that with the 8.73 mg we are trying to measure — and ask whether noise was ever the problem.

Three more terms, and one of them is not small

QUANTISATION

$$\text{LSB} / \sqrt{12}$$

$$\text{A: } 0.061/3.46 = 0.018$$

$$\text{B: } 0.488/3.46 = 0.141$$

Both utterly negligible against 8.73 mg.

OFFSET

calibrated out

$$\text{A: } \pm 40 \text{ mg} \rightarrow 0$$

$$\text{B: } \pm 10 \text{ mg} \rightarrow 0$$

A single bench calibration removes this — at one temperature.

OFFSET DRIFT

$$\text{TC} \times \Delta T$$

$$\text{A: } 0.5 \times 20 = 10.0$$

$$\text{B: } 0.1 \times 20 = 2.0$$

Calibrated at 20 °C, used 0–40 °C, so $\Delta T = \pm 20$ °C.

Part A drifts by 10 mg. The entire quantity we are trying to measure is 8.73 mg.

The error budget

Root-sum-square, because the terms are independent

Term	Part A	Part B	Where it came from
Noise ($100 \mu\text{g}/\sqrt{\text{Hz}} \times \sqrt{50 \text{ Hz}}$)	0.707	0.707	datasheet $\times$ your bandwidth
Quantisation ($\text{LSB}/12$)	0.018	0.141	resolution and full scale
Offset drift ($\text{TC} \times \Delta T$)	10.000	2.000	temp. coefficient $\times$ 20 °C
TOTAL (RSS)	10.02	2.13	$\sqrt{(\text{sum of squares})}$
AS A TILT ANGLE	0.574°	0.122°	$\text{asin}(\text{mg} / 1000)$
AGAINST $\pm 0.5^\circ$	FAILS	passes, 4× margin	the requirement

All values in mg unless stated. Bandwidth 50 Hz, calibrated at 20 °C, used over 0–40 °C.

One term dominates. It was on page nine, and it was not on either front page.

Now the part on your bench

ISM330DHCX · the same budget, run twice from its own datasheet

Term	using TYP	using MAX	from DS13012 Rev 6
Noise density	60 $\mu\text{g}/\sqrt{\text{Hz}}$	100 $\mu\text{g}/\sqrt{\text{Hz}}$	both printed, same row
Noise, $\times \sqrt{50 \text{ Hz}}$	0.424	0.707	your bandwidth
Quantisation, LSB/ $\sqrt{12}$	0.018	0.018	16-bit, $\pm 2 \text{ g}$
Offset drift TC	$\pm 0.1 \text{ mg}/^\circ\text{C}$	$\pm 0.5 \text{ mg}/^\circ\text{C}$	both printed, same row
Drift, $\times 20^\circ\text{C}$	2.000	10.000	0–40 $^\circ\text{C}$, cal at 20 $^\circ\text{C}$
TOTAL (RSS)	2.05	10.02	$\sqrt{(\text{sum of squares})}$
AS A TILT ANGLE	0.117°	0.574°	$\text{asin}(\text{mg}/1000)$
AGAINST $\pm 0.5^\circ$	passes, 4× margin	FAILS	the requirement

One part number. One datasheet. Two columns — and the requirement sits between them.

Which part do you specify for a $\pm 0.5^\circ$ tilt measurement?

A Part A — eight times finer resolution, and cheaper

B Part B

C This cannot be decided from front pages alone

The right FIRST answer was C — you did not have enough information. The right SECOND answer, after twelve minutes of arithmetic, is B.

Part A resolves eight times finer than Part B — and cannot do the job.

Part B resolves 0.028° of tilt: eighteen times finer than the $\pm 0.5^\circ$ we need. Resolution was never the deciding variable. It only looked like it was, because it was the number on the front page.

Your turn: three candidates

Teams of three · 8 minutes · commit to one

Criterion	Weight	A	B	C
Meets $\pm 0.5^\circ$ over 0–40 °C	pass/fail	fail	pass	pass
Total error as an angle	high	0.574°	0.122°	0.06°
Unit price at 500 off	medium	€1.80	€4.20	€11.50
Current draw	medium	0.15 mA	0.18 mA	0.90 mA
Library already exists	LOW	yes	no	no

PART C

16-bit, ± 2 g

60 $\mu\text{g}/\sqrt{\text{Hz}}$

offset ± 5 mg

TC ± 0.05 mg/°C

€11.50, 0.9 mA

Write one sentence: “We specify Part ____ because ____” — naming the **DOMINANT ERROR TERM**, not the headline.

Part C is the trap in the other direction: technically the best part, six times the price, and it buys no capability the requirement asks for. Over-specifying is also an engineering failure.

New problem. Water level in an outdoor tank, 2 m deep (≈ 20 kPa full scale), to ± 20 mm, water 5–35 °C. P: total error band ± 0.25 % FS over 0–50 °C. Q: ± 0.05 % FS at 25 °C, temperature coefficient not specified.

- A** Sensor Q — five times better accuracy
- B** Sensor P — its error is bounded over the whole temperature range; Q's is specified at one point only
- C** Either — the temperature coefficient can be measured later
- D** Sensor Q, calibrated at 25 °C before installation

Different measurand, different error term. Same question underneath.

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A total error band over the operating range beats a headline accuracy at one temperature. An unspecified coefficient is not a small error — it is an unbounded one.

The course's selection rule

Quoted from your own semester plan

“Avoid selecting a part only because an Arduino library exists; students must still see the underlying configuration and data path.”

Part A had a library. Part A cannot do the job.

What to choose on instead — parts that expose the engineering:

selectable range and ODR

documented bandwidth and
noise

self-test

data-ready interrupt

FIFO where relevant

an accessible register map

a complete datasheet

Four things to keep

01 Turn the requirement into a number before you read any datasheet.

0.5° became 8.73 mg, and 8.73 mg decided everything that followed.

02 Find the dominant term. Everything else is decoration.

10 mg and 0.7 mg combine to 10.02 mg. The small term never mattered.

03 Resolution is not accuracy, and neither is precision.

Systematic → calibration. Random → bandwidth. Drift → component choice.

04 A number without its conditions is not a number.

typ at 25 °C is not a guarantee, and an unspecified coefficient is unbounded.

Exit ticket

Anonymous · hand it in at the door

1 · COMPUTE

A sensor gives $200 \mu\text{g}/\sqrt{\text{Hz}}$.
You set a 100 Hz bandwidth.

State the RMS noise.

2 · JUDGEMENT

Name the one specification you
will look for first, for the rest
of your career, before the
headline number.

3 · MUDDIEST POINT

What is the one thing today
you are least sure about?

One sentence.

Question 1 is the one I will actually grade myself on. If the room cannot do it, I taught it badly.

Tuesday: Laboratory 1

Datasheet to data — bring the pre-lab sheet, signed

You already found number 5 on the hunt: the device-identification register.
On Tuesday you read it off real silicon — and prove your sensor is the sensor you think it is.

AND THEN LECTURE 3

From physical quantity to trustworthy samples — sampling, aliasing, quantisation, and the ADC configuration that killed the motor.